

Locators in Selenium – Java

Corporate Training | Real-Time
Examples | Best Practices

What are Locators?

- Locators identify web elements on a web page
- Selenium uses By class to locate elements
- Required for click, sendKeys, getText operations

Types of Locators

- ID, Name, ClassName
- TagName
- LinkText & PartialLinkText
- CSS Selector
- XPath

ID Locator – Best Practice

- HTML: `<input id='username'>`
- Java:
`driver.findElement(By.id("username")).sendKeys("admin");`
- ✓ Fastest and most reliable

Name Locator

- HTML: `<input name='email'>`
- Java: `driver.findElement(By.name("email"))`
- ⚠ Not always unique

Class Name Locator

- HTML: `<button class='btn login'>Login</button>`
- Java: `driver.findElement(By.className("btn"))`
-  Cannot use multiple class values

LinkText & PartialLinkText

- HTML: `<a>Login</a>`
- `By.linkText("Login")`
- `By.partialLinkText("Log")`

CSS Selector – Modern & Fast

- #id, .class, tag
- input[name='email']
- button[type='submit']
- driver.findElement(By.cssSelector("#password"))

XPath – Powerful Locator

- `//input[@id='username']`
- `//button[text()='Login']`
- `//input[contains(@id,'user')]`

Advanced XPath Axes

- following-sibling
- ancestor
- descendant
- Interview favourite

XPath vs CSS Selector

- CSS is faster
- XPath supports text & parent traversal
- Use CSS when possible

Handling Dynamic Elements

- Use contains()
- Use starts-with()
- Avoid absolute XPath

Shadow DOM – Concept

- Elements hidden inside shadow root
- Normal XPath/CSS cannot access directly
- Requires JavaScript execution

Real-Time Example

- Login button inside form
- `//form[@id='login']//button[@type='submit']`

Locator Priority – Interview

- 1. ID
- 2. Name
- 3. CSS Selector
- 4. XPath
- 5. Class Name

Common Mistakes

- Using absolute XPath
- Relying on dynamic classes
- Ignoring waits

Best Practices

- Keep locators simple
- Use Page Object Model
- Centralize locators

Summary

- Locators are foundation of Selenium
- Choose stability over complexity
- Practice real applications